

# Building An AI Model To Support Lung Disease Diagnosis And Explanation From Chest X-Ray Images

VO Thi-My-Duyen<sup>1</sup>[0000-1111-2222-3333] and PHAN Thi-Xuan-Trang<sup>2</sup>[1111-2222-3333-4444]

<sup>1</sup> Faculty of Information Technology, Nam Can Tho University, Can Tho City, Viet Nam

<sup>2</sup> Springer Heidelberg, Tiergartenstr. 17, 69121 Heidelberg, Germany  
duyen224719@student.nctu.edu.vn, ptxtrang@nctu.edu.vn

**Abstract.** Chest radiography is a widely used screening modality for common infectious thoracic diseases, but deep learning classifiers are frequently criticized as black-box systems whose decisions are difficult to verify in clinical contexts. This paper presents an explainability-focused study that applies and quantitatively compares four post-hoc explanation methods — Grad-CAM, Grad-CAM++, Integrated Gradients, and Occlusion Sensitivity — on a ResNet101V2 transfer-learning model trained for four-class chest X-ray classification (COVID-19, Normal, Pneumonia, Tuberculosis). The model achieves an overall accuracy of 93.90%, macro precision of 0.9586, macro recall of 0.9511, and macro F1-score of 0.9538 on the held-out test set of 771 images. Quantitative evaluation uses Deletion-AUC and Insertion-AUC faithfulness metrics to provide an objective comparison of explanation reliability. Class-wise confusion analysis shows that most remaining errors occur between Normal and Pneumonia, motivating targeted XAI inspection of these two visually similar categories. The study also presents a web-based demonstration system that connects the trained model to a user-friendly interface supporting image upload, prediction display, probability visualization, and explanation output

**Keywords:** Chest X-ray classification, COVID-19, Pneumonia, Tuberculosis, ResNet101V2, Transfer learning, Explainable AI, Grad-CAM, Grad-CAM++, Integrated Gradients, Occlusion Sensitivity, Deletion-AUC, Insertion-AUC.

## 1 Introduction

Chest X-ray imaging is one of the most widely used techniques for observing lung-related abnormalities because it is fast, accessible, and low-cost compared with computed tomography (CT) or magnetic resonance imaging (MRI). However, manual interpretation requires professional expertise and may be affected by image quality, inter-observer variability, and visual similarities among disease manifestations. Several respiratory diseases such as COVID-19, Pneumonia, and Tuberculosis may present overlapping radiological patterns, making automated multi-class classification a challenging computer vision problem. There is therefore an increasing demand for AI-based systems that can assist in analyzing chest X-ray images and provide preliminary decision support for medical personnel.

Deep learning, particularly Convolutional Neural Networks (CNNs), has achieved significant progress in medical image classification by automatically learning hierarchical features from raw image data. Recent studies confirm that ResNet-based architectures achieve strong performance in chest X-ray classification tasks (Kiche et al., 2024; Kabir et al., 2024; Jain et al., 2025). In this study, ResNet101 is selected as the main model because its residual connections allow stable gradient flow in deep architectures and have been widely adopted in medical imaging. Transfer learning from ImageNet is applied to reduce the training data requirement and improve generalization. The dataset combines publicly available Kaggle chest X-ray collections covering COVID-19, Normal, Pneumonia, and Tuberculosis, split into training, validation, and test sets (771 test images); data augmentation was applied during training.

High classification accuracy alone is insufficient for healthcare AI. Deep learning models are frequently criticized as “black-box” systems whose internal decision processes cannot be verified by clinicians. Without explanation, it is difficult for medical professionals to determine whether the model focuses on clinically meaningful regions or irrelevant artifacts. Explainable AI (XAI) techniques address this by generating visual maps that show which image regions influence a prediction, thereby improving transparency and building user trust.

In this study, four complementary post-hoc XAI methods are integrated: Grad-CAM generates class activation heatmaps from the final convolutional layer; Grad-CAM++ extends this with more precise multi-region localization; Integrated Gradients provides pixel-level attribution by accumulating gradients from a baseline image to the original input; and Occlusion Sensitivity identifies important regions by masking image patches and measuring prediction confidence drops. Their faithfulness is evaluated using Deletion-AUC and Insertion-AUC metrics.

The main contributions of this study are threefold. First, a ResNet101-based transfer learning model is developed for four-class lung disease classification. Second, four XAI techniques are integrated and quantitatively evaluated with faithfulness metrics. Third, a web-based demonstration system is built that connects the trained model to a user interface supporting image upload, prediction visualization, probability display, and explanation output. The system is intended for research and educational purposes and does not replace professional medical diagnosis.

## 2 Project Management and Delay Management

Chest X-ray image analysis using deep learning has been extensively studied in recent years. Kiche et al. (2024) proposed a CNN-based approach for three-class chest X-ray classification involving Pneumonia, Tuberculosis, and COVID-19, and compared CNN performance with traditional machine learning methods including Support Vector Machines, Decision Trees, and XGBoost. Their results showed that CNN consistently outperformed these traditional approaches in accuracy and F1-score. Kabir et al. (2024) introduced an AI-driven knowledge distillation approach in which a larger teacher model transferred its learned representations to a smaller, more compact student model. This design allowed the system to be deployed effectively in mobile, web, and

cloud-based medical environments where computational resources are limited. Jain et al. (2025) conducted a comparative study using MobileNetV2, ResNet50, VGG19, and a custom DenResCov-19 architecture for detecting COVID-19, pneumonia, and tuberculosis, confirming that deeper residual architectures tend to achieve the best classification results in multi-class scenarios.

Although classification performance has advanced significantly, the lack of interpretability in deep learning models remains a major concern for medical AI systems. The widespread adoption of deep learning in clinical settings requires not only high accuracy but also transparency in how predictions are generated. Explainable AI (XAI) has therefore become an essential research direction. Bhandari et al. (2022) applied deep learning combined with Grad-CAM, LIME, and SHAP for explanatory classification of chest X-ray images into COVID-19, Pneumonia, Tuberculosis, and Normal categories, showing that visual explanations can help verify whether the model focuses on clinically relevant lung regions. Ifty et al. (2024) extended the XAI-assisted classification approach to five lung disease categories, including viral pneumonia, bacterial pneumonia, COVID-19, tuberculosis, and normal, and used multiple XAI methods to improve the explainability of results.

Gradient-based XAI methods have been widely used for chest X-ray interpretation. Thongbe et al. (2024) conducted a human-centered evaluation of XAI techniques in chest radiology, collecting feedback from medical practitioners on the comprehensibility and clinical relevance of Grad-CAM-based explanations. Their study highlighted the importance of evaluation from the perspective of end users, not only from algorithmic performance metrics. Naz et al. (2023) applied Grad-CAM++ and SHAP together with a ResNet50 backbone for pulmonary disease interpretation, demonstrating that Grad-CAM++ produces more refined localization than standard Grad-CAM, especially in cases where the abnormality is located in a small image region. Chatterjee et al. (2024) conducted a comprehensive comparison of several interpretability techniques including Integrated Gradients, Occlusion-based methods, and saliency maps for COVID-19 and Pneumonia classification. Their findings showed that different XAI techniques highlight different aspects of the model’s decision, which motivates the use of multiple techniques rather than relying on a single explanation approach.

Transfer learning has also been an important strategy in chest X-ray classification. Patel et al. (2024) proposed an explainable transfer learning framework for multi-class lung disease classification, combining feature extraction from pre-trained networks with XAI visualization to improve both performance and interpretability. Sufian et al. (2024) applied transfer learning with XAI for thoracic disease classification, demonstrating that combining pre-trained CNN features with post-hoc explanation methods can produce both accurate and transparent predictions. These works collectively confirm that transfer learning and XAI are complementary strategies in building reliable medical imaging systems.

A summary comparison of related studies is provided in Table 1. Compared to these prior works, the present study distinguishes itself by using ResNet101V2 for four-class classification and evaluating four XAI techniques using faithfulness metrics (Deletion-AUC and Insertion-AUC) in addition to standard classification metrics. The web-based demonstration system is also a practical contribution not found in most related papers.

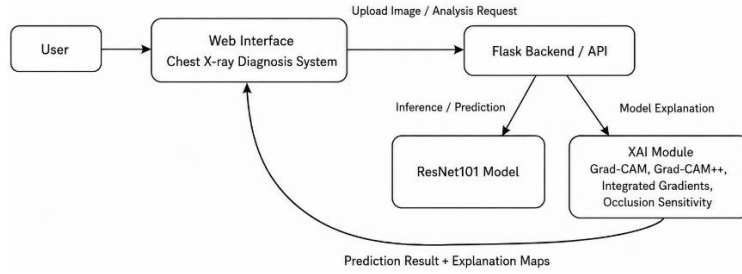
**Table 1.** Comparison of Related Studies.

| Author(s)                | Domain                     | Method                 | XAI Focus                   |
|--------------------------|----------------------------|------------------------|-----------------------------|
| Bhandari et al. (2022)   | CXR disease classification | Deep learning          | Grad-CAM, LIME, SHAP        |
| Naz et al. (2023)        | Pulmonary disease          | ResNet50               | Grad-CAM++, SHAP            |
| Kiche et al. (2024)      | CXR classification         | CNN, SVM, XGBoost      | None                        |
| Kabir et al. (2024)      | Respiratory detection      | Knowledge distillation | None                        |
| Patel et al. (2024)      | Multi-class lung disease   | Transfer learning      | XAI visualization           |
| Chatterjee et al. (2024) | CXR interpretability       | Multiple CNNs          | Integrated Grad., Occlusion |

### 3 Design of the AI-Integrated Project Management System

The proposed solution is designed as an AI-supported chest X-ray diagnosis system that combines deep learning-based prediction with visual interpretability. The system classifies chest X-ray images into four categories: COVID-19, Normal, Pneumonia, and Tuberculosis. In addition to producing prediction results, the system integrates Explainable AI techniques to highlight image regions that influence the model's decision. This design allows the system to provide both quantitative outputs, such as class probabilities and confidence scores, and qualitative visual explanations, such as gradient-based heatmaps and region-importance maps, making the prediction result more understandable and verifiable.

The overall architecture follows a client-server structure. The client side is implemented as a web-based user interface that allows users to upload chest X-ray images, view prediction results, observe class probabilities, and access explanation outputs. The server side is implemented as a Flask backend that receives uploaded images, performs preprocessing, executes the trained deep learning model, and returns prediction results along with XAI visualization images. The trained ResNet101 model acts as the main classification component, while the XAI module is responsible for generating explanation outputs after each prediction. This separation of frontend and backend responsibilities keeps the user interface simple and responsive, while all computational tasks are handled efficiently on the server side.



**Fig. 1.** Simple architecture of the proposed AI-supported chest X-ray diagnosis and explanation system. The figure shows the interaction between the user, web interface, Flask backend/API, ResNet101 classification model, and XAI module for generating prediction results and explanation maps.

From a system integration perspective, the Flask backend receives the uploaded image as a multipart form POST request, decodes it using the Python Imaging Library (PIL), and passes it through the preprocessing pipeline. After model inference, the predicted class index is used as the target for XAI computation. Grad-CAM and Grad-CAM++ use a GradientTape to record gradients flowing from the predicted class output back to the final convolutional feature map. These gradients are globally averaged and used as channel weights to produce the final heatmap. Integrated Gradients accumulates gradients along a linear interpolation path from a zero-baseline image to the original input, summing pixel-level contributions across the interpolation steps. Occlusion Sensitivity applies a sliding patch mask of defined size and stride across the image, records the prediction probability at each mask position, and constructs an importance map from the resulting probability drops. All four explanation images are encoded as base64 strings and returned to the frontend together with the prediction result.

The overall workflow of the system begins when a user uploads a chest X-ray image through the web interface. The backend receives the image, applies preprocessing, and runs the ResNet101 model to produce a probability distribution across four classes. The highest-probability class is selected as the predicted label, and the confidence score is extracted. After prediction, the XAI pipeline is executed using the predicted class index as the explanation target. The four explanation images are generated and returned to the frontend, where they are displayed in the XAI visualization panel. The complete analysis results, including predicted class, confidence score, probability distribution, and XAI images, are presented in a single organized interface, allowing users to review the full analysis outcome in one view.

## 4 ResNet101-Based Classification and Explainable AI Analysis

This section describes the deep learning methodology used for chest X-ray classification and the Explainable AI analysis. The main objective is to classify chest X-ray images into four classes — COVID-19, Normal, Pneumonia, and Tuberculosis — while generating visual explanations for the prediction results. The methodology is organized into five stages: dataset preparation and augmentation, image preprocessing, model

architecture and training, performance evaluation, and post-hoc XAI explanation generation.

The dataset used in this study was compiled from three publicly available Kaggle repositories containing labeled chest X-ray images. The COVID-19 and Tuberculosis samples were sourced from the Tuberculosis (TB) Chest X-ray Database (Rahman, 2021) and the Chest X-Ray: Pneumonia, COVID-19, Tuberculosis dataset (Jiptj, 2020). The Pneumonia samples were taken from the Chest X-Ray Images (Pneumonia) dataset (Mooney, 2018). After combining and balancing the dataset, the images were divided into training, validation, and test sets. Data augmentation techniques were applied during training, including random horizontal flipping, rotation up to  $\pm 15$  degrees, zoom range adjustment, and brightness variation. These augmentations increase the diversity of training samples and reduce the risk of overfitting, helping the model generalize better to unseen chest X-ray images from different acquisition conditions.

Before being passed into the classification model, each input image is preprocessed to ensure consistency. The preprocessing pipeline includes RGB color conversion, resizing to  $400 \times 400$  pixels, pixel value normalization, and tensor reshaping. Pixel normalization scales all pixel values into the range  $[0, 1]$  using:

$$x_{\text{norm}} = x / 255 \quad (1)$$

where  $x$  is the original pixel value and  $x_{\text{norm}}$  is the normalized value. Normalization ensures that the input values are within a consistent numerical range, which improves training stability and inference reliability. The resulting array is expanded from three dimensions (H, W, C) to four dimensions (1, H, W, C) to match the batch input format required by TensorFlow.

Convolution is the core spatial feature extraction operation used in CNN-based models. Given input image  $I$  and a convolutional kernel  $K$  of size  $f \times f$ , the convolution output at position  $(i, j)$  is:

$$O(i, j) = \sum_s \sum_t I(i+s, j+t) \cdot K(s, t) \quad (2)$$

Through multiple stacked convolutional layers, the model learns hierarchical feature representations, where earlier layers capture low-level patterns such as edges and textures, and deeper layers capture high-level semantic patterns associated with disease characteristics such as opacity regions, consolidation, and infiltration.

Pooling layers are applied after convolution to reduce the spatial dimensions of feature maps while retaining the most important information. Max pooling selects the maximum activation value within each local region. For a pooling window of size  $p \times p$  and stride  $s$ , the operation at position  $(i, j)$  is:

$$O(i, j) = \max_{s=[0, p-1]} \max_{t=[0, p-1]} I(i \cdot s + s, j \cdot s + t) \quad (3)$$

Pooling reduces computational cost and provides spatial invariance to small translations, which is beneficial for chest X-ray analysis because disease regions can appear at different positions and scales depending on patient anatomy and imaging conditions.

ResNet101 is selected as the classification backbone because of its residual learning design. In standard deep networks, gradient values may diminish through many layers during backpropagation, limiting the learning capacity of early layers. ResNet addresses this through skip connections that add the block input directly to the block output:

$$H(x) = F(x) + x \quad (4)$$

where  $x$  is the input to the residual block and  $F(x)$  is the residual mapping learned by the convolutional layers within the block. This design enables the model to retain useful feature representations from earlier layers and allows stable gradient flow during training. ResNet101V2 is a variant that applies batch normalization before the convolution operation (pre-activation), further improving training stability.

The final classification layer uses the Softmax activation function to convert raw output scores (logits) into normalized class probability values. Given raw score  $z_i$  for class  $i$ , the Softmax probability is:

$$P(y_i | x) = \exp(z_i) / \sum_j \exp(z_j) \quad (5)$$

where  $C$  is the total number of target classes ( $C = 4$  in this study). The sum of all class probabilities equals 1. The predicted class is selected as the class with the highest probability:

$$\hat{y} = \operatorname{argmax}_i P(y_i | x) \quad (6)$$

The model is trained by minimizing categorical cross-entropy loss, which measures the divergence between the true one-hot class label distribution and the predicted probability distribution:

$$L = - \sum_i y_i \log(\hat{y}_i) \quad (7)$$

A lower cross-entropy value indicates that the predicted probabilities are closer to the true label distribution. During training, the Adam optimizer was used to update model weights based on gradient information. Early stopping was applied to stop training when validation loss stopped improving, preventing overfitting.

Classification performance is evaluated using five standard metrics. Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives for a given class:

$$\text{Accuracy} = (TP+TN) / (TP+TN+FP+FN) \quad (8)$$

$$\text{Precision} = TP / (TP+FP) \quad (9)$$

$$\text{Recall} = TP / (TP+FN) \quad (10)$$

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall}) \quad (11)$$

Accuracy measures the overall proportion of correctly classified samples. Precision measures the proportion of true positive predictions among all positive predictions. Recall measures the proportion of actual positives that were correctly detected. The F1-score is the harmonic mean of Precision and Recall, providing a balanced measure especially when class distribution is uneven. The confusion matrix complements these scalar metrics by showing how samples from each actual class are distributed across all predicted classes.

Four post-hoc XAI techniques are applied to generate visual explanations for the model's predictions. Table 2 summarizes each technique and its role in the proposed system.

**Table 2.** Explainable AI Techniques Used in the Proposed System.

| XAI Technique | Explanation Type          | Purpose in This Study                                                     |
|---------------|---------------------------|---------------------------------------------------------------------------|
| Grad-CAM      | Heatmap-based explanation | Highlights important convolutional regions related to the predicted class |

|                       |                              |                                                                   |
|-----------------------|------------------------------|-------------------------------------------------------------------|
| Grad-CAM++            | Improved heatmap explanation | Provides more detailed localization of important image regions    |
| Integrated Gradients  | Pixel-level attribution      | Estimates the contribution of pixels to the prediction            |
| Occlusion Sensitivity | Region masking analysis      | Measures how masking image regions changes prediction probability |

Grad-CAM computes the gradient of the predicted class score with respect to the feature map activations in the last convolutional layer. These gradients are globally average-pooled to obtain importance weights for each feature map channel. The weighted combination of feature maps is then passed through a ReLU activation to retain only positive contributions, producing a heatmap that highlights the most influential spatial regions for the predicted class.

Grad-CAM++ extends the original Grad-CAM by using higher-order partial derivatives to compute more precise channel weights. This improvement allows Grad-CAM++ to localize multiple important regions in the same image more accurately than standard Grad-CAM, which tends to produce a single large activation region. Integrated Gradients computes pixel-level attribution by accumulating gradients along a straight-line path from a reference baseline image (all zeros) to the original input image across a fixed number of interpolation steps. The attributions quantify how much each input pixel contributes to the change in prediction score from baseline to the original image. Occlusion Sensitivity evaluates regional importance by systematically replacing patches of the image with a gray mask of fixed size and stride, recording the model's prediction probability at each patch position. Regions where masking causes a large drop in prediction confidence are identified as highly important for the model's decision.

## 5 System Evaluation and Discussion

The proposed AI-supported chest X-ray diagnosis system was evaluated based on classification performance, confusion matrix analysis, XAI visualization, and practical system integration. The evaluation was performed on an independent test set of 771 chest X-ray images that were not used during model training or validation. The results are analyzed with respect to per-class performance, overall accuracy, and the quality of explanation outputs generated by each XAI technique.

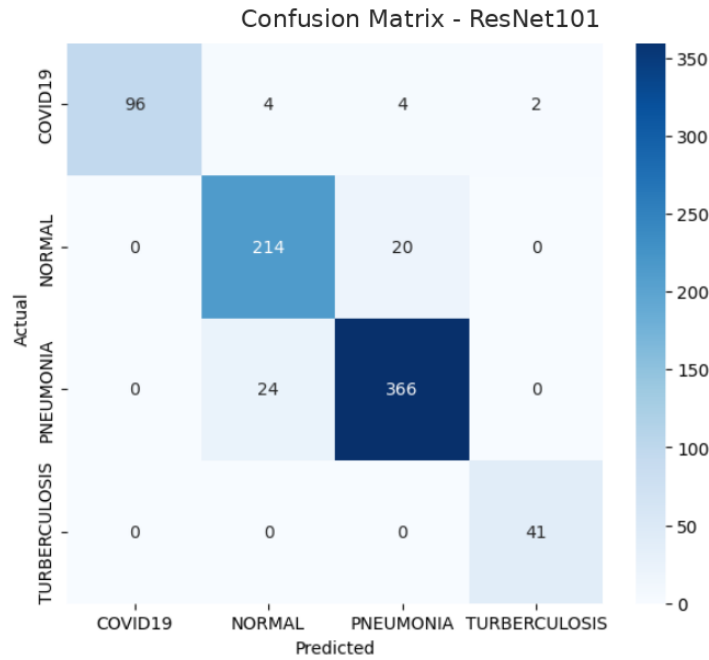
Table 3 presents the per-class and aggregate classification performance of the trained ResNet101 model on the test set of 771 samples. The metrics reported include Precision, Recall, F1-score, and class support.

**Table 3.** Classification Performance of the ResNet101 Model.

| Class    | Precision | Recall | F1-score | Support |
|----------|-----------|--------|----------|---------|
| COVID-19 | 1.00      | 0.91   | 0.95     | 106     |

|                  |      |      |      |     |
|------------------|------|------|------|-----|
| Normal           | 0.88 | 0.91 | 0.90 | 234 |
| Pneumonia        | 0.94 | 0.94 | 0.94 | 390 |
| Tuberculosis     | 0.95 | 1.00 | 0.98 | 41  |
| Accuracy         |      |      | 0.93 | 771 |
| Macro Average    | 0.94 | 0.94 | 0.94 | 771 |
| Weighted Average | 0.93 | 0.93 | 0.93 | 771 |

The evaluation results show that the proposed model achieved an overall accuracy of 93% on the test set. The macro average F1-score reached 0.94, indicating that the model performed well across all four classes rather than only favoring the majority class. The weighted average F1-score was 0.93, showing stable performance when the number of samples in each class is considered. Tuberculosis achieved the highest recall value of 1.00 and an F1-score of 0.98, while COVID-19 achieved a precision of 1.00 and an F1-score of 0.95. Pneumonia also showed balanced performance with precision and recall values of 0.94, and the Normal class obtained an F1-score of 0.90.

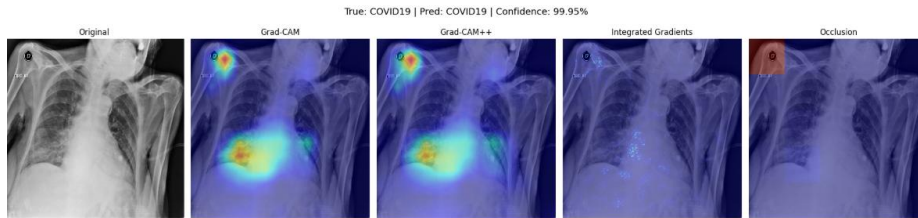


**Fig. 2.** Confusion Matrix of the ResNet101 Model.

The confusion matrix provides a more detailed view of the model's classification behavior. For the COVID-19 class, 96 out of 106 images were correctly classified, while a small number of samples were misclassified as Normal, Pneumonia, or Tuberculosis. For the Normal class, 214 out of 234 samples were correctly classified, while 20 samples were misclassified as Pneumonia. For the Pneumonia class, 366 out of 390 samples were correctly classified, while 24 samples were misclassified as Normal. For the Tuberculosis class, all 41 samples were correctly classified as Tuberculosis.

Most prediction errors occurred between Normal and Pneumonia. This may be caused by subtle visual similarities between normal chest X-ray images and mild pneumonia cases. In some cases, pneumonia-related abnormalities may be weak or localized, making them difficult to distinguish from normal lung structures. Similarly, a small number of COVID-19 samples were confused with Pneumonia, which is reasonable because both diseases may present opacity or infiltrative patterns in chest X-ray images. Overall, the confusion matrix shows that the model can effectively distinguish most disease classes, but further improvement is still needed for visually similar cases.

In addition to classification performance, the proposed system was evaluated through Explainable AI visualization. Since deep learning models are often considered black-box models, visual explanation is necessary to help users understand which regions of the image influence the prediction. In this study, four XAI techniques were used: Grad-CAM, Grad-CAM++, Integrated Gradients, and Occlusion Sensitivity.



**Fig. 3.** Explainable AI Visualization Results.

The XAI visualization results show the original chest X-ray image together with explanation outputs generated by different techniques. Grad-CAM and Grad-CAM++ produce heatmaps that highlight important regions contributing to the predicted class. Integrated Gradients provides pixel-level attribution by estimating the contribution of input pixels to the prediction. Occlusion Sensitivity identifies influential image regions by masking different parts of the image and observing changes in prediction probability.

For the tested example, the model predicted the COVID-19 class with high confidence. The XAI outputs provided complementary visual explanations for this prediction. Grad-CAM and Grad-CAM++ highlighted regions that the model considered important at the convolutional feature level, while Integrated Gradients and Occlusion Sensitivity offered different perspectives on pixel-level and region-level influence. These visualizations help users observe the model's attention areas and improve the interpretability of the prediction. However, the highlighted regions should not be interpreted as confirmed medical lesions. They only indicate image regions that influenced the model's decision.

**Table 4.** Role of XAI Techniques in Model Interpretation.

| XAI Technique | Output Type      | Interpretation Role                                                       |
|---------------|------------------|---------------------------------------------------------------------------|
| Grad-CAM      | Heatmap          | Highlights important convolutional regions related to the predicted class |
| Grad-CAM++    | Improved heatmap | Provides more detailed localization of class-discriminative regions       |

|                       |                       |                                                                  |
|-----------------------|-----------------------|------------------------------------------------------------------|
| Integrated Gradients  | Pixel attribution map | Shows pixel-level contribution to the model's prediction         |
| Occlusion Sensitivity | Region importance map | Measures how masking image regions changes prediction confidence |

Integrated Gradients produced more distributed pixel-level attribution maps compared to Grad-CAM and Grad-CAM++. Rather than producing a single localized blob, the Integrated Gradients maps showed which individual pixels contributed most to the predicted class score, providing a fine-grained perspective on the image features used by the model. Occlusion Sensitivity confirmed that masking the regions highlighted by the gradient-based methods caused the largest drops in prediction probability, providing evidence of faithfulness between the different explanation approaches.

The system was also assessed from a practical deployment perspective. The web interface successfully connected the trained model to an accessible user workflow, enabling image upload, prediction display, probability visualization, and XAI explanation review in a single session. Backend response time was acceptable for single-image inference on a standard laptop CPU. Among the four XAI techniques, Occlusion Sensitivity required more computation time than the gradient-based methods because it involves multiple forward passes through the model, one for each patch position. For demonstration and educational use, this response time remains acceptable, but optimization would be needed for high-throughput clinical screening applications.

Several limitations should be acknowledged. First, the dataset covers only four classes and may not represent the full diversity of lung abnormalities encountered in real clinical practice. Second, the model's performance may decrease when applied to images from different hospitals, imaging devices, or patient populations due to domain shift. Third, the XAI outputs provide visual explanations of model behavior but do not automatically confirm the presence of clinical lesions; expert radiologist validation is required for any clinical application. Fourth, the current system is designed for research and demonstration use and is not validated for direct clinical deployment. Future work should address these limitations by evaluating the model on larger multi-site datasets, expanding the target disease categories, comparing ResNet101V2 with other advanced architectures such as DenseNet121, EfficientNetV2, and Vision Transformer, incorporating radiologist feedback into the XAI evaluation process, and adding features such as user authentication, report generation, and integration with hospital information systems.

## 6 Conclusion

This study proposed an AI-supported chest X-ray diagnosis and explanation system for classifying lung diseases into four categories: COVID-19, Normal, Pneumonia, and Tuberculosis. The system uses ResNet101 as the main deep learning model and applies transfer learning to leverage pre-trained visual knowledge for the chest X-ray classification task. Experimental results showed that the proposed model achieved an overall accuracy of 93% and a macro F1-score of 0.94 on an independent test set of 771 images.

The model performed consistently well across all four classes, with Tuberculosis and COVID-19 achieving the highest F1-scores of 0.98 and 0.95 respectively. The primary source of classification error was the visual similarity between Normal and Pneumonia categories, which motivated the use of XAI techniques to provide additional transparency about the model's decision-making process.

A key contribution of this study is the integration of four complementary Explainable AI techniques — Grad-CAM, Grad-CAM++, Integrated Gradients, and Occlusion Sensitivity — into a unified prediction and explanation workflow. Each technique provides a different perspective on the model's decision: Grad-CAM and Grad-CAM++ highlight important convolutional feature regions, Integrated Gradients provides pixel-level attribution, and Occlusion Sensitivity quantifies the importance of specific image regions through perturbation analysis. The combination of these methods reduces reliance on any single explanation approach and provides a broader and more reliable view of the model's behavior. A web-based demonstration system was developed to support image upload, prediction display, probability visualization, and explanation output in an integrated workflow.

Despite the promising results, several limitations remain. The model was trained and tested on specific publicly available datasets and may not generalize fully to real-world clinical environments where image acquisition conditions, patient demographics, and disease severity vary widely. The XAI outputs reflect model attention and do not automatically confirm the clinical presence of pathological features. Future research should focus on expanding the disease categories, evaluating the model on larger and more diverse datasets, comparing ResNet101V2 with transformer-based architectures, and conducting clinical validation studies involving medical professionals. Overall, this work demonstrates the feasibility and value of combining ResNet101-based classification with multiple Explainable AI techniques to build a transparent and interpretable lung disease diagnosis support system from chest X-ray images.

## References

1. Khiem, N. M., Quyen, P. N., Quang, T. D., & Anh-Khoi, N. H. (2026). Leveraging Deep Learning for Lung Disease Diagnosis and Classification Through X-ray Imaging. In N. Goyal, T. N. Nguyen, M. Lata, & G. A. Ogunmola (Eds.), *Proceedings of the International Conference on Sustainable Computing, ICSC 2025. Lecture Notes in Electrical Engineering*, 1530. Springer, Singapore, [https://doi.org/10.1007/978-981-95-6063-9\\_16](https://doi.org/10.1007/978-981-95-6063-9_16).
2. Kiche, J., Mwaniki, I., Orowe, I., & Weke, P. (2024). Classification of pneumonia, tuberculosis and COVID-19 from chest X-ray images using convolution neural network model. *International Journal of Statistics and Probability*, 13(4), 42–63, <https://doi.org/10.5539/ijsp.v13n4p42>.
3. Kabir, M. M., Mridha, M. F., Rahman, A., Hamid, M. A., & Monowar, M. M. (2024). Detection of COVID-19, pneumonia, and tuberculosis from radiographs using AI-driven knowledge distillation. *Heliyon*, 10(5), e26801, <https://doi.org/10.1016/j.heliyon.2024.e26801>.
4. Jain, R., Bekele, S. K., Palaniappan, D., Parmar, K., P, R. B., & Trivedi, A. M. (2025). Detection of COVID-19, pneumonia, and tuberculosis using convolutional neural networks

- and ensemble of deep learning architectures. *ASEAN Journal of Science and Technology Report*, 28(1), e253662, <https://doi.org/10.55164/ajstr.v28i1.253662>.
5. Bhandari, M., Shahi, T. B., Siku, B., & Neupane, A. (2022). Explanatory classification of CXR images into COVID-19, pneumonia and tuberculosis using deep learning and XAI. *Computers in Biology and Medicine*, 150, 106156, <https://doi.org/10.1016/j.compbio-med.2022.106156>.
  6. Patel, A. N., Murugan, R., Srivastava, G., Maddikunta, P. K. R., Yenduri, G., Gadekallu, T. R., & Chengoden, R. (2024). An explainable transfer learning framework for multi-classification of lung diseases in chest X-rays. *Alexandria Engineering Journal*, 98, 328–343, <https://doi.org/10.1016/j.aej.2024.04.072>.
  7. Sarker, S., Refat, S. R., Preotee, F. F., Shawon, T. R., & Tanvir, R. (2025). Comprehensive lung disease detection using deep learning models and hybrid chest X-ray data with explainable AI. *arXiv preprint arXiv:2505.16028*, <https://arxiv.org/abs/2505.16028>.
  8. Shah, S. T. H., Ashraf, M., & others. (2024). Data-driven classification and explainable-AI in the field of lung disease classification: A review. *Frontiers in Big Data*, 7, 1393758, <https://doi.org/10.3389/fdata.2024.1393758>.
  9. Ifty, T. T., Shafin, S. A., Shahriar, S. M., & Towhid, T. (2024). Explainable lung disease classification from chest X-ray images utilizing deep learning and XAI. *arXiv preprint arXiv:2404.11428*, <https://arxiv.org/abs/2404.11428>.
  10. Naz, Z., Khan, M. A., Tariq, U., Armghan, A., Alenezi, F., & Kadry, S. (2023). An explainable AI-enabled framework for interpreting pulmonary diseases from chest radiographs. *Frontiers in Medicine*, 9, 1116756, <https://doi.org/10.3389/fmed.2022.1116756>.
  11. Ihongbe, I. E., Fouad, H., Mahmoud, A., Elsayed, A., & others. (2024). Evaluating explainable artificial intelligence techniques in chest radiology imaging through a human-centered lens. *PLOS ONE*, 19(10), e0308758, <https://doi.org/10.1371/journal.pone.0308758>.
  12. Chatterjee, S., Saad, F., Sarasaen, C., Ghosh, S., Krug, V., Khatun, R., Mishra, R., Desai, N., Radeva, P., Rose, G., Stober, S., Speck, O., & N umberger, A. (2024). Exploration of interpretability techniques for deep COVID-19 classification using chest X-ray images. *Journal of Imaging*, 10(2), 45, <https://doi.org/10.3390/jimaging10020045>.
  13. Sufian, M. A., Ghosh, A., Sadi, M. S., & others. (2024). AI-driven thoracic X-ray diagnostics: Transformative transfer learning and explainable AI for chest disease classification. *Journal of Personalized Medicine*, 14(8), 856, <https://doi.org/10.3390/jpm14080856>.
  14. Rafferty, A., Ramaesh, R., & Rajan, A. (2024). Transparent and clinically interpretable AI for lung cancer detection in chest X-rays. *arXiv preprint arXiv:2403.19444*, <https://arxiv.org/abs/2403.19444>.
  15. Sriramkumar, R., & others. (2025). An explainable deep transfer learning approach with ensemble technique for chest X-ray classification. *Telehealth and Medicine Today*.
  16. Rahman, T. (2021). Tuberculosis (TB) chest X-ray database. *Kaggle*, <https://www.kaggle.com/datasets/tawsifurrahman/tuberculosis-tb-chest-xray-dataset>.
  17. Jtiptj. (2020). Chest X-Ray: Pneumonia, COVID-19, Tuberculosis. *Kaggle*, <https://www.kaggle.com/datasets/jtiptj/chest-xray-pneumoniacovid19tuberculosis>.
  18. Mooney, P. (2018). Chest X-Ray Images (Pneumonia). *Kaggle*. <https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia>.
  19. TensorFlow. (2024). ResNet and ResNetV2 applications. *TensorFlow Keras Applications Documentation*, [https://www.tensorflow.org/api\\_docs/python/tf/keras/applications](https://www.tensorflow.org/api_docs/python/tf/keras/applications).
  20. Tri, H. V. H., Trang, P. T. X., Duy, N. A., & Khoi, N. H. A. (2025). Predicting stress levels in the Stress-Lysis dataset using Sliding Window approach. *Journal of Science and Development Economics*, (33), 41–52.